1. **Aim/Overview of the practical:**
   (A) Write a program for Support Vector Machine using python.

   (B) Write a program based on logistic regression and make the sigmoid graph using sigmoid function.

2. **Objective of the practical:**

   (A) To classify data into distinct categories using Support Vector Machine (SVM).
   (B) To perform binary classification using Logistic Regression and visualize the sigmoid function.

3. **Code for experiment/practical:**

(A)

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.svm import SVC

# Load data
data = pd.read_csv("diabetes_dataset.csv")
X = data[['Glucose', 'BMI']]
y = data['Outcome']

# Train SVM
svm_model = SVC(kernel='linear')
svm_model.fit(X, y)

# Create grid
xx, yy = np.meshgrid(
    np.linspace(X['Glucose'].min()-5, X['Glucose'].max()+5, 100),
    np.linspace(X['BMI'].min()-5, X['BMI'].max()+5, 100)
)
Z = svm_model.predict(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)
```

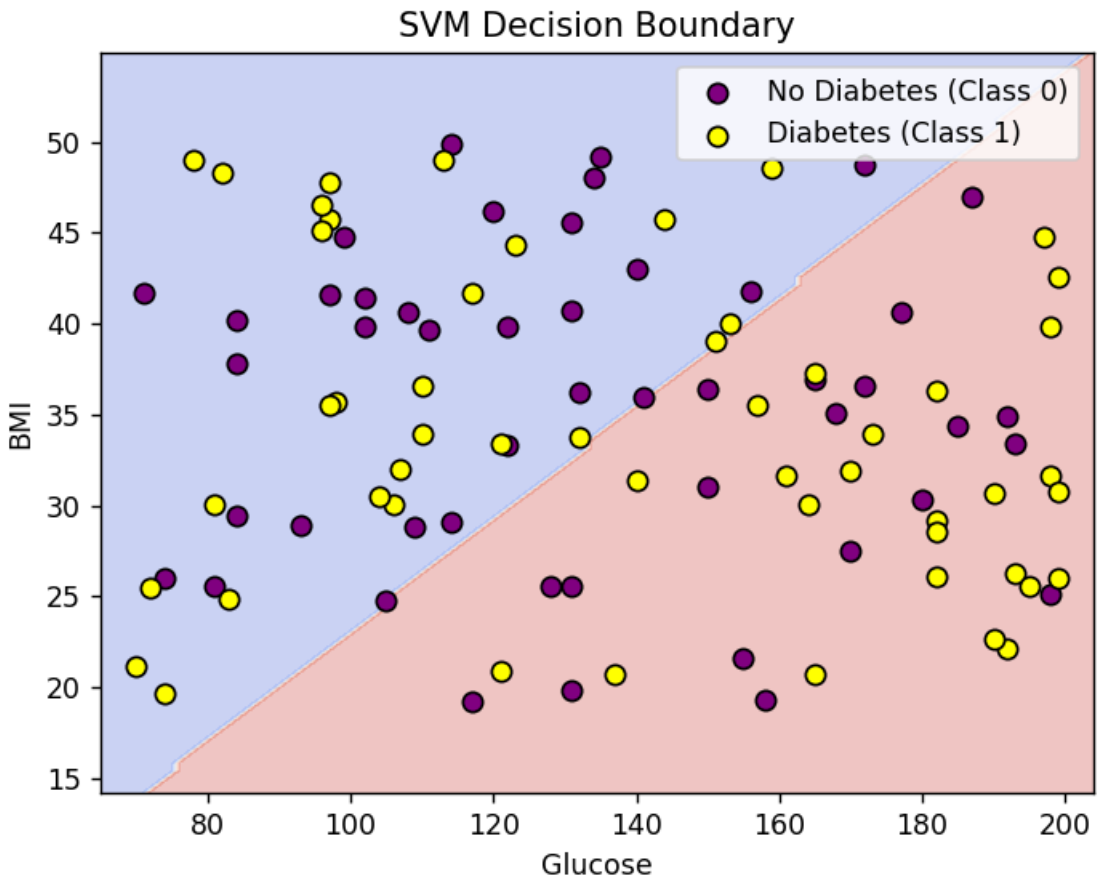
```
# Plot decision boundary
plt.contourf(xx, yy, Z, alpha=0.3, cmap=plt.cm.coolwarm)

# Plot data points with labels
for class_value, color, label in zip([0, 1], ['purple', 'yellow'], ['No Diabetes (Class 0)', 'Diabetes (Class 1)']):
    plt.scatter(
        X[y == class_value]['Glucose'],
        X[y == class_value]['BMI'],
        color=color, edgecolor='k', s=50,
        label=label
    )

plt.xlabel('Glucose')
plt.ylabel('BMI')
plt.title('SVM Decision Boundary')
plt.legend()
plt.show()
```

(B)

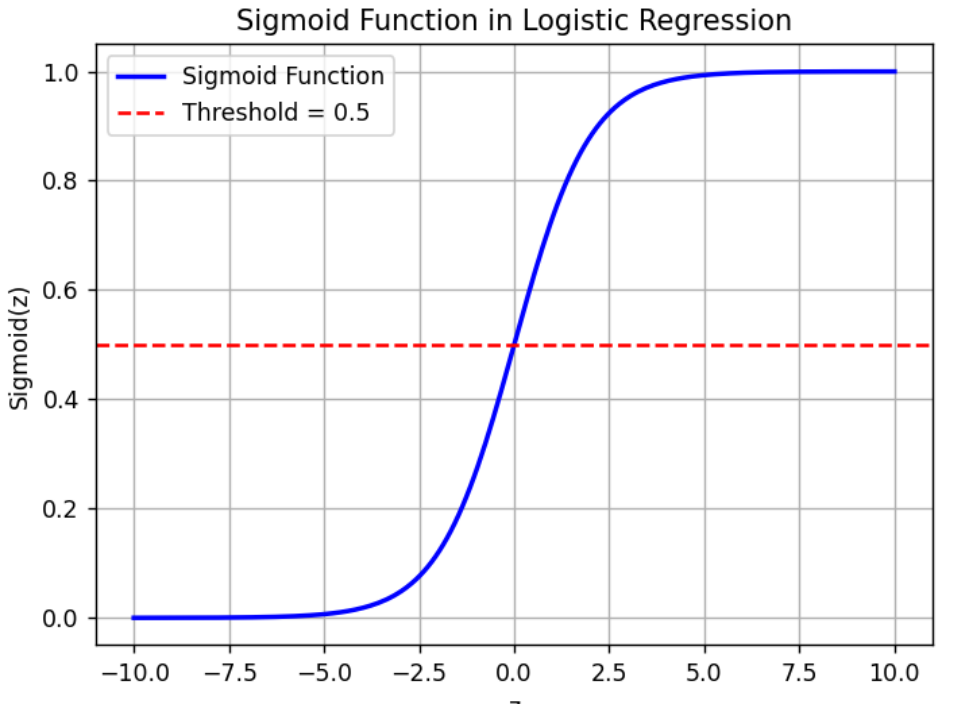
```
import numpy as np
import matplotlib.pyplot as plt

# Step 1: Define sigmoid function
def sigmoid(z):
    return 1 / (1 + np.exp(-z))

# Step 2: Generate values
z = np.linspace(-10, 10, 200)  # x-axis values
s = sigmoid(z)                 # apply sigmoid

# Step 3: Plot sigmoid curve
plt.plot(z, s, color='blue', linewidth=2, label="Sigmoid Function")
plt.axhline(0.5, color='red', linestyle="--", label="Threshold = 0.5")
plt.xlabel("z")
plt.ylabel("Sigmoid(z)")
plt.title("Sigmoid Function in Logistic Regression")
plt.legend()
plt.grid(True)
plt.show()
```

4. **Output:**

(A)


SVM Decision Boundary
No Diabetes (Class 0)
Diabetes (Class 1)
BMI
Glucose
50
45
40
35
30
25
20
15
80
100
120
140
160
180
200

(B)



5. Learning Outcomes (What I have learnt):

1. Understood SVM and its hyperplane classification.
2. Learned Logistic Regression for binary classification.
3. Understood sigmoid function for probability mapping.